Welcome to the
Web Application Development with Python L-4
Training Program

Md. Shakil - Assessor & Trainer - National Skill Development Authority (NSDA)

Lecture

08

Python List

Md. Shakil
Assessor & Trainer
National Skill Development Authority

Python List

- A **list** is a built-in data structure in Python that allows you to store **multiple items** in a **single variable**.
- Lists are **ordered**
- **mutable** (can be changed)
- **allow duplicate values**.
- Lists are created using square brackets:

```
studentlist = [“rohim”, 100, True, “Korim”, 5.5]
```

Why Use Lists?

Lists are used to:

- Store multiple values in a single variable
- Access elements using index
- Modify, add, or remove elements
- Perform operations on collections of data

Creating a List

Lists are created using **square brackets [ ]**.

```python
# Creating a list
my_list = [10, 20, 30, 40, 50]

# Lists can store mixed data types
mixed_list = [1, "hello", 3.14, True]

# Empty list
empty_list = []
```

Access List Element

List elements are accessed using **index numbers** (starting from 0). The index of first item is **0**, the index of second item is **1**, and so on.

List →	['Python',	'Swift',	'C++']
Index →	0	1	2

Access List Element

```python
languages = ['Python', 'Swift', 'C++']

# access the first element
print('languages[0] =', languages[0])

# access the third element
print('languages[2] =', languages[2])
```

Negative Indexing

In Python, a list can also have negative indices. The index of the last element is **-1**, the second last element is **-2** and so on.

	['Python',	'Swift',	'C++']
Index →	0	1	2
Negative Index →	-3	-2	-1

Changing List Items

Lists are mutable, so values can be changed.

```
languages = ["Python", "Swift", "C++"]
languages[1] = "JavaScript"
print(languages)
```

List Length

Use the **len()** function to find the number of elements

```
languages = ["Python", "Swift", "C++"]
print(len(languages))
```

Adding Elements to a List

append(item) Method

Adds a single item to the end of the list

```
fruits = ['apple', 'banana']
fruits.append('orange')
print(fruits)  # ['apple', 'banana', 'orange']
```

insert(index, item) Method

Inserts an item at the specified index

```
fruits = ['apple', 'banana']
fruits.insert(1, 'grape')
print(fruits)  # ['apple', 'grape', 'banana']
```

extend(list) Method

Add the element of a list to the end of the current list

```
fruits = ['apple', 'banana']
fruits.extend(['orange', 'kiwi'])
print(fruits)  # ['apple', 'banana', 'orange',
'kiwi']
```

Removing Elements from a List

remove(item) Method

Removes the first occurrence of a value

```
fruits = ['apple', 'banana', 'apple']
fruits.remove('apple')
print(fruits)  # ['banana', 'apple']
```

pop([index]) Method

Removes the element at the specified position

Note: If we not provide index it’s remove the last element

```
fruits = ['apple', 'banana', 'cherry']
fruits.pop()        # Removes 'cherry'
fruits.pop(0)       # Removes 'apple'
print(fruits)       # ['banana']
```

clear() Method

Removes all items from the list.

```
fruits = ['apple', 'banana']
fruits.clear()
print(fruits)  # []
```

del Keyword

Remove specific index or a variable.

```
languages = ["Python", "Swift", "C++"]
del languages[1]
print(languages)
```

Using for Loop

```python
languages = ["Python",'C++', "Swift", "C++"]
for lang in languages:
    print(lang)
```

count() Method

Returns the number of elements with the specified value.

```
numbers = [1, 2, 2, 3, 2]
print(numbers.count(2))  # 3
```

copy() method

Duplicate or copy a list using this method.

```
nums = [5, 2, 8, 1]
new_list = nums.copy()
```

Thanks All